

Reactome Analysis Result

03 January, 2025

Introduction

This report summarizes the pathway analysis result created by the **Reactome Analysis Service**. For more information, visit <https://www.reactome.org>.

In this reports, pathways found differentially expressed at an adjusted p-value ≤ 0.05 are considered to be significantly regulated.

Analysis Overview

- Number of datasets analysed: 4
- Reactome version: 91
- Disease pathways were included

Dataset summaries

PL2-3_2DG_vs_Ctrl:

- 2581 pathways
- 10240 fold changes for genes / proteins

R848_2DG_vs_Ctrl:

- 2582 pathways
- 10289 fold changes for genes / proteins

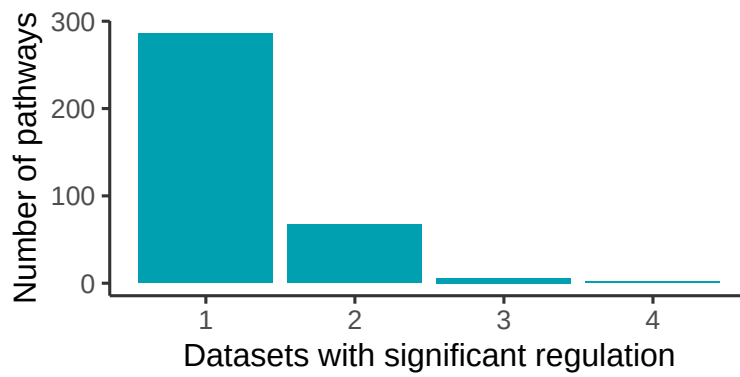
PL2-3_vs_R848:

- 2582 pathways
- 10288 fold changes for genes / proteins

PL2-3_2DG_vs_R848_2DG:

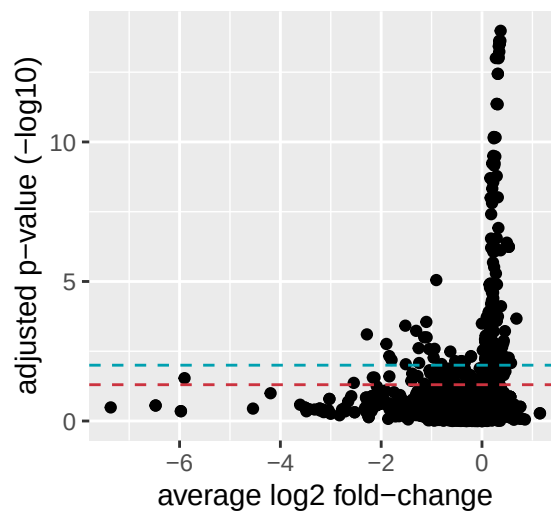
- 2581 pathways
- 10267 fold changes for genes / proteins

Number of datasets the different pathways were found to be significantly regulated in:

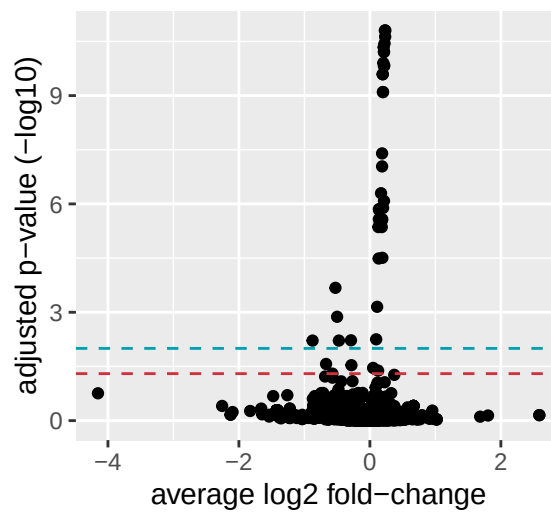


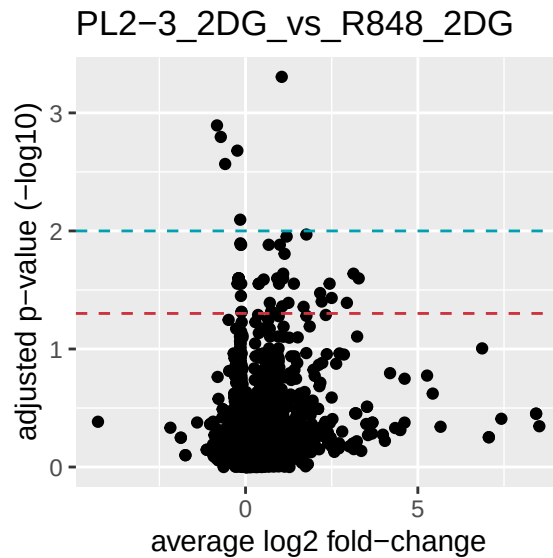
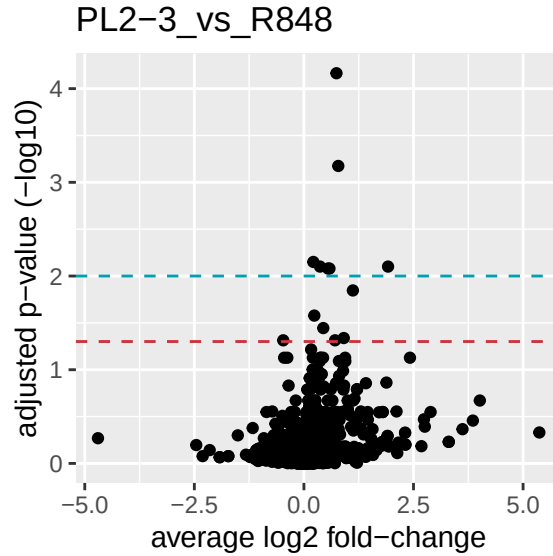
Pathway analysis

PL2-3_2DG_vs_Ctrl



R848_2DG_vs_Ctrl

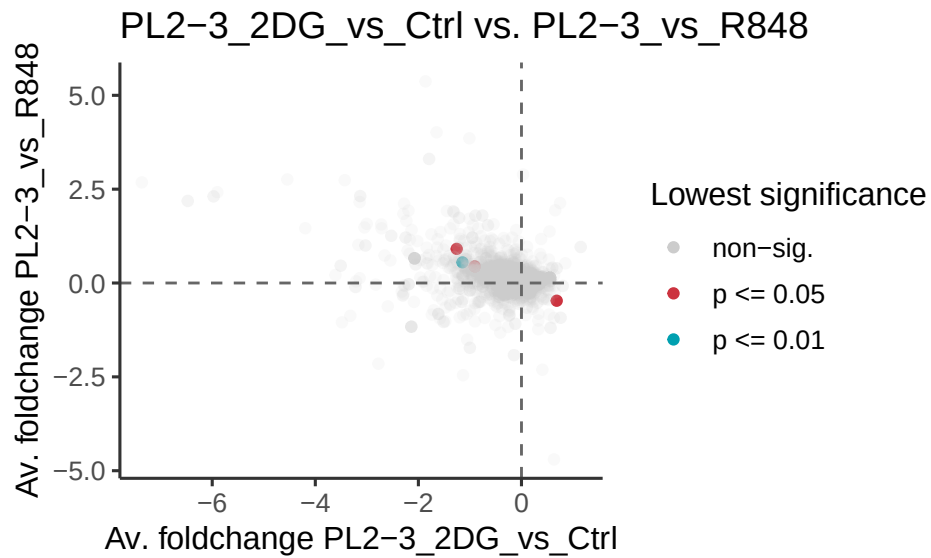
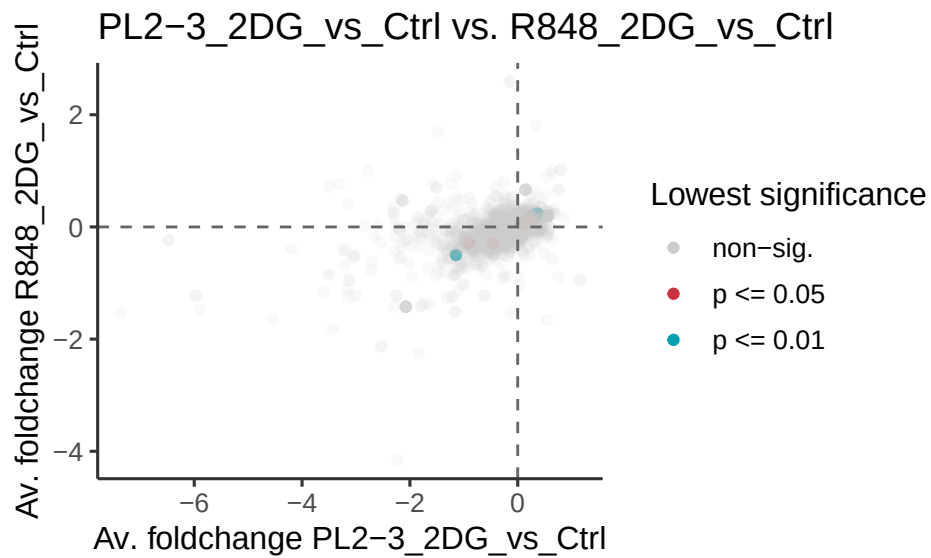


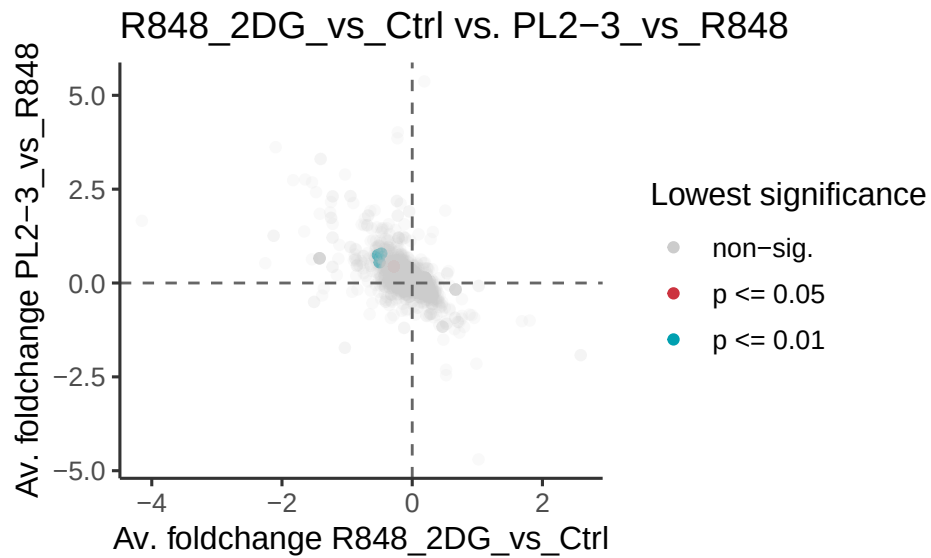
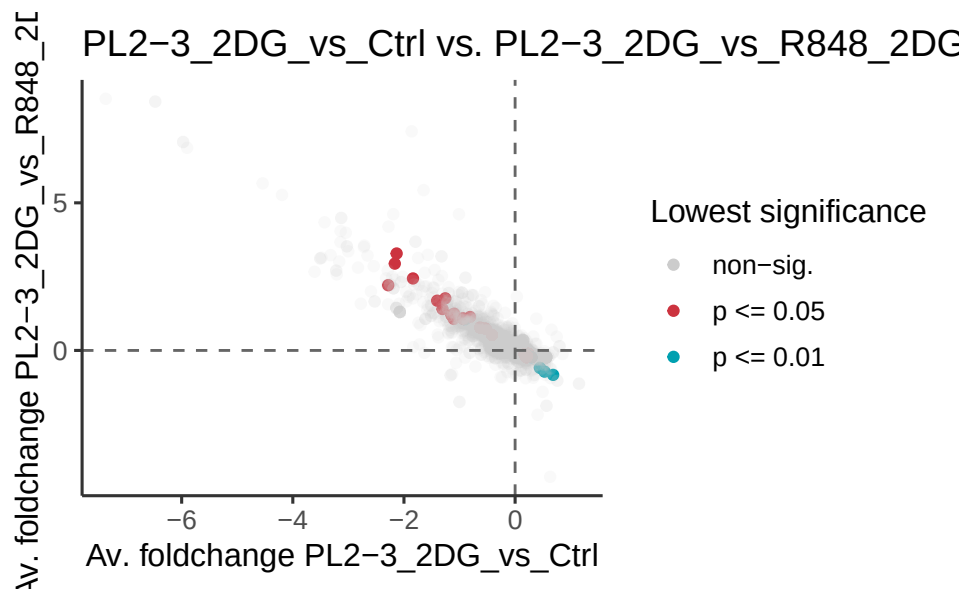


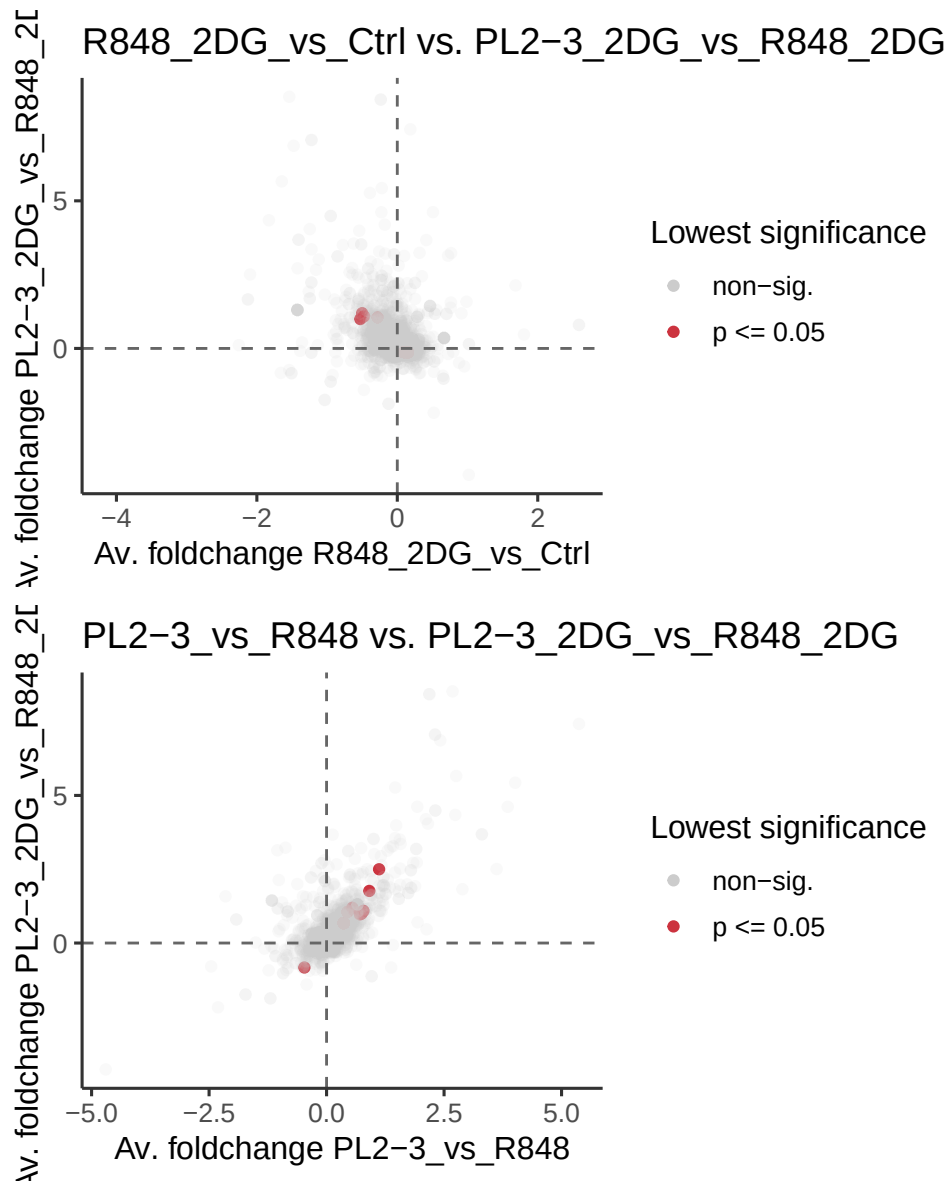
These volcano plots summarize the pathway results for every dataset. Every point is one pathway. The x-axis represents the average fold-change of all genes / proteins within that pathway. The y-axis represents the p-value where “higher” values are more significant ($-\log_{10}$ transformation). The red line represents $p = 0.05$, and the blue line $p = 0.01$.

Pathway correlation

Correlation between all datasets (if more than one). Every point represents one pathway, with the x-axis showing the average fold-change of one dataset and the y-axis the average fold-change of the other dataset. Only pathways that were observed in both datasets are shown. Points are colored based on the lowest observed significance.







Similar regulated pathways

Pathways that show the same direction on all of the datasets. The direction of regulation is only assessed for significantly regulated pathways.

Viral mRNA Translation (R-HSA-192823)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Up	Up	Down	Down
FDR	< 0.001	< 0.001	0.984	0.073
Av. FC	0.37	0.234	-0.023	-0.157
N Genes	74	74	74	74

Peptide chain elongation (R-HSA-156902)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Up	Up	Down	Down
FDR	< 0.001	< 0.001	0.983	0.109
Av. FC	0.364	0.236	-0.023	-0.15
N Genes	74	74	74	74

Eukaryotic Translation Termination (R-HSA-72764)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Up	Up	Up	Down
FDR	< 0.001	< 0.001	0.996	0.078
Av. FC	0.341	0.205	0.003	-0.153
N Genes	78	78	78	77

Formation of a pool of free 40S subunits (R-HSA-72689)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Up	Up	Up	Down
FDR	< 0.001	< 0.001	0.997	0.136
Av. FC	0.347	0.224	-0.022	-0.143
N Genes	86	86	86	86

Selenocysteine synthesis (R-HSA-2408557)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Up	Up	Down	Down
FDR	< 0.001	< 0.001	0.994	0.097
Av. FC	0.351	0.219	-0.02	-0.149
N Genes	78	78	78	78

Eukaryotic Translation Elongation (R-HSA-156842)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Up	Up	Down	Down
FDR	< 0.001	< 0.001	0.983	0.118
Av. FC	0.352	0.229	-0.024	-0.144
N Genes	77	77	77	77

Nonsense Mediated Decay (NMD) independent of the Exon Junction Complex (EJC) (R-HSA-975956)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Up	Up	Down	Down
FDR	< 0.001	< 0.001	0.997	0.115
Av. FC	0.333	0.211	0.001	-0.141
N Genes	80	80	80	79

Response of EIF2AK4 (GCN2) to amino acid deficiency (R-HSA-9633012)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Up	Up	Down	Down
FDR	< 0.001	< 0.001	0.984	0.109
Av. FC	0.344	0.215	0.016	-0.111
N Genes	86	86	86	86

GTP hydrolysis and joining of the 60S ribosomal subunit (R-HSA-72706)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Up	Up	Down	Down
FDR	< 0.001	< 0.001	0.984	0.145
Av. FC	0.33	0.215	-0.026	-0.139
N Genes	97	97	97	97

Regulation of expression of SLITs and ROBOs (R-HSA-9010553)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Up	Up	Up	Down
FDR	< 0.001	< 0.001	0.872	0.099
Av. FC	0.275	0.138	0.001	-0.148
N Genes	132	132	132	131

L13a-mediated translational silencing of Ceruloplasmin expression (R-HSA-156827)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Up	Up	Down	Down
FDR	< 0.001	< 0.001	0.996	0.17
Av. FC	0.328	0.215	-0.022	-0.133
N Genes	96	96	96	96

Eukaryotic Translation Initiation (R-HSA-72613)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Up	Up	Up	Down
FDR	< 0.001	< 0.001	0.986	0.168
Av. FC	0.315	0.2	-0.018	-0.131
N Genes	104	104	104	104

Cap-dependent Translation Initiation (R-HSA-72737)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Up	Up	Up	Down
FDR	< 0.001	< 0.001	0.986	0.168
Av. FC	0.315	0.2	-0.018	-0.131
N Genes	104	104	104	104

Nonsense-Mediated Decay (NMD) (R-HSA-927802)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Up	Up	Down	Down
FDR	< 0.001	< 0.001	0.923	0.109
Av. FC	0.293	0.196	-0.022	-0.134
N Genes	100	100	100	99

Nonsense Mediated Decay (NMD) enhanced by the Exon Junction Complex (EJC) (R-HSA-975957)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Up	Up	Down	Down
FDR	< 0.001	< 0.001	0.923	0.109
Av. FC	0.293	0.196	-0.022	-0.134
N Genes	100	100	100	99

SRP-dependent cotranslational protein targeting to membrane (R-HSA-1799339)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Up	Up	Up	Down
FDR	< 0.001	< 0.001	0.94	0.168
Av. FC	0.312	0.186	-0.003	-0.126
N Genes	96	96	96	96

Influenza Viral RNA Transcription and Replication (R-HSA-168273)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Up	Up	Down	Down
FDR	< 0.001	< 0.001	0.997	0.136
Av. FC	0.27	0.171	-0.018	-0.115
N Genes	118	118	118	118

Immunoregulatory interactions between a Lymphoid and a non-Lymphoid cell (R-HSA-198933)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Down	Down	Up	Up
FDR	< 0.001	0.188	0.305	0.052
Av. FC	-1.52	-0.258	0.532	1.767
N Genes	87	87	87	87

Co-inhibition by PD-1 (R-HSA-389948)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Down	Up	Down	Up
FDR	0.002	0.913	0.932	0.064
Av. FC	-1.896	0.051	0.039	1.862
N Genes	20	20	20	20

ECM proteoglycans (R-HSA-3000178)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Down	Down	Up	Up
FDR	0.003	0.21	0.323	0.065
Av. FC	-1.047	-0.366	0.456	1.105
N Genes	31	32	32	32

Phosphorylation of CD3 and TCR zeta chains (R-HSA-202427)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Down	Up	Down	Up
FDR	0.005	0.652	0.73	0.108
Av. FC	-1.839	0.127	-0.074	1.763
N Genes	19	19	19	19

IRE1alpha activates chaperones (R-HSA-381070)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Down	Down	Up	Up
FDR	0.005	0.171	0.778	0.27
Av. FC	-0.219	-0.111	0.009	0.118
N Genes	49	49	49	49

XPB1(S) activates chaperone genes (R-HSA-381038)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Down	Down	Up	Up
FDR	0.005	0.218	0.811	0.275
Av. FC	-0.215	-0.103	-0.001	0.112
N Genes	47	47	47	47

GPCR ligand binding (R-HSA-500792)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Down	Down	Up	Up
FDR	0.005	0.294	0.377	0.078
Av. FC	-0.958	-0.303	0.443	1.11
N Genes	159	163	163	163

Activation of Matrix Metalloproteinases (R-HSA-1592389)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Down	Down	Up	Up
FDR	0.007	0.776	0.664	0.052
Av. FC	-1.794	-0.212	0.751	2.329
N Genes	17	17	17	17

Glycosaminoglycan metabolism (R-HSA-1630316)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Down	Down	Up	Up
FDR	0.007	0.891	0.927	0.111
Av. FC	-0.404	-0.104	0.117	0.416
N Genes	84	84	84	84

Selenoamino acid metabolism (R-HSA-2408522)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Up	Up	Up	Down
FDR	< 0.001	< 0.001	0.984	0.298
Av. FC	0.264	0.187	0.003	-0.068
N Genes	96	97	97	97

SARS-CoV-1 modulates host translation machinery (R-HSA-9735869)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Up	Up	Down	Down
FDR	< 0.001	< 0.001	0.932	0.258
Av. FC	0.325	0.215	-0.03	-0.139
N Genes	33	33	33	33

Formation of the ternary complex, and subsequently, the 43S complex (R-HSA-72695)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Up	Up	Down	Down
FDR	< 0.001	< 0.001	0.983	0.403
Av. FC	0.292	0.202	-0.028	-0.116
N Genes	48	48	48	48

Interferon alpha/beta signaling (R-HSA-909733)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Down	Down	Up	Up
FDR	0.013	0.006	0.687	0.564
Av. FC	-0.459	-0.289	0.191	0.366
N Genes	56	56	56	56

Initial triggering of complement (R-HSA-166663)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Down	Down	Up	Up
FDR	0.234	0.006	0.453	0.947
Av. FC	-1.171	-0.875	0.661	0.956
N Genes	17	17	17	17

Binding and Uptake of Ligands by Scavenger Receptors (R-HSA-2173782)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Down	Down	Up	Up
FDR	0.28	0.027	0.213	0.659
Av. FC	-1.067	-0.669	0.649	1.05
N Genes	25	25	25	25

Detoxification of Reactive Oxygen Species (R-HSA-3299685)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Up	Down	Up	Up
FDR	0.808	0.703	0.007	0.183
Av. FC	-0.109	-0.057	0.215	0.267
N Genes	29	29	29	29

Neutrophil degranulation (R-HSA-6798695)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Down	Down	Up	Up
FDR	0.226	0.691	0.008	0.013
Av. FC	-0.418	-0.123	0.367	0.674
N Genes	396	400	400	400

Formyl peptide receptors bind formyl peptides and many other ligands (R-HSA-444473)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Down	Down	Up	Up
FDR	0.361	0.369	0.008	0.109
Av. FC	-1.447	-0.643	1.919	2.726
N Genes	6	6	6	6

Basigin interactions (R-HSA-210991)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Down	Down	Up	Up
FDR	0.296	0.21	0.008	0.078
Av. FC	-0.474	-0.346	0.582	0.71
N Genes	20	20	20	20

Fibronectin matrix formation (R-HSA-1566977)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Down	Down	Up	Up
FDR	0.113	0.527	0.014	0.037
Av. FC	-2.02	-0.638	1.117	2.499
N Genes	4	4	4	4

Nuclear events mediated by NFE2L2 (R-HSA-9759194)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Up	Down	Up	Up
FDR	0.293	0.705	0.026	0.782
Av. FC	-0.178	-0.145	0.237	0.271
N Genes	78	78	78	78

Localization of the PINCH-ILK-PARVIN complex to focal adhesions (R-HSA-446343)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Down	Down	Up	Up
FDR	0.185	0.754	0.049	0.028
Av. FC	-0.392	-0.132	0.71	0.971
N Genes	2	2	2	2

MET interacts with TNS proteins (R-HSA-8875513)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Down	Up	Up	Up
FDR	0.093	0.982	0.279	0.023
Av. FC	-1.746	0.098	1.299	3.133
N Genes	5	5	5	5

The role of Nef in HIV-1 replication and disease pathogenesis (R-HSA-164952)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Down	Down	Up	Up
FDR	0.144	0.957	0.412	0.025
Av. FC	-0.801	0.046	0.071	0.92
N Genes	27	27	27	27

alpha-linolenic (omega3) and linoleic (omega6) acid metabolism (R-HSA-2046104)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Down	Down	Up	Up
FDR	0.268	0.886	0.127	0.028
Av. FC	-0.189	-0.082	0.278	0.384
N Genes	10	10	10	10

alpha-linolenic acid (ALA) metabolism (R-HSA-2046106)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Down	Down	Up	Up
FDR	0.268	0.886	0.127	0.028
Av. FC	-0.189	-0.082	0.278	0.384
N Genes	10	10	10	10

CHL1 interactions (R-HSA-447041)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Down	Down	Up	Up
FDR	0.166	0.853	0.081	0.034
Av. FC	-1.633	-0.416	0.945	2.165
N Genes	6	6	6	6

Differently regulated pathways

Mitochondrial translation initiation (R-HSA-5368286)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Up	Up	Up	Down
FDR	< 0.001	0.447	0.655	0.028
Av. FC	0.236	0.087	0.022	-0.126
N Genes	87	87	87	87

Mitochondrial translation termination (R-HSA-5419276)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Up	Up	Up	Down
FDR	< 0.001	0.436	0.551	0.049
Av. FC	0.236	0.09	0.029	-0.117
N Genes	87	87	87	87

Mitochondrial translation elongation (R-HSA-5389840)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Up	Up	Up	Down
FDR	< 0.001	0.518	0.501	0.048
Av. FC	0.236	0.084	0.032	-0.118
N Genes	87	87	87	87

Extracellular matrix organization (R-HSA-1474244)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Down	Down	Up	Up
FDR	< 0.001	0.029	0.036	< 0.001
Av. FC	-0.907	-0.282	0.442	1.051
N Genes	164	166	166	166

Degradation of the extracellular matrix (R-HSA-1474228)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Down	Down	Up	Up
FDR	< 0.001	0.218	0.35	0.025
Av. FC	-1.104	-0.399	0.395	1.069
N Genes	62	63	63	63

Peptide ligand-binding receptors (R-HSA-375276)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Down	Down	Up	Up
FDR	0.001	0.082	0.213	0.028
Av. FC	-1.306	-0.438	0.545	1.403
N Genes	73	75	75	75

Translocation of ZAP-70 to Immunological synapse (R-HSA-202430)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Down	Up	Down	Up
FDR	0.001	0.733	0.836	0.04
Av. FC	-2.283	0.144	-0.062	2.21
N Genes	16	16	16	16

Integrin cell surface interactions (R-HSA-216083)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Down	Down	Up	Up
FDR	0.001	0.001	0.008	0.011
Av. FC	-1.145	-0.501	0.55	1.193
N Genes	49	49	49	49

Class A/1 (Rhodopsin-like receptors) (R-HSA-373076)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Down	Down	Up	Up
FDR	0.001	0.249	0.349	0.041
Av. FC	-1.098	-0.321	0.456	1.245
N Genes	121	125	125	125

Interleukin-10 signaling (R-HSA-6783783)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Down	Down	Up	Up
FDR	0.002	0.229	0.046	0.011
Av. FC	-1.256	-0.4	0.909	1.766
N Genes	34	34	34	34

Non-integrin membrane-ECM interactions (R-HSA-3000171)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Down	Down	Up	Up
FDR	0.003	0.686	0.369	0.023
Av. FC	-0.942	-0.183	0.335	1.092
N Genes	47	47	47	47

Regulation of Insulin-like Growth Factor (IGF) transport and uptake by Insulin-like Growth Factor Binding Proteins (IGFBPs) (R-HSA-381426)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Down	Down	Up	Up
FDR	0.003	0.361	0.428	0.049
Av. FC	-0.628	-0.253	0.402	0.777
N Genes	59	59	59	59

Post-translational protein phosphorylation (R-HSA-8957275)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Down	Down	Up	Up
FDR	0.007	0.361	0.305	0.048
Av. FC	-0.532	-0.261	0.471	0.741
N Genes	55	55	55	55

Lysosomal oligosaccharide catabolism (R-HSA-8853383)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Down	Down	Up	Up
FDR	0.007	0.847	0.655	0.026
Av. FC	-0.417	-0.067	0.177	0.528
N Genes	4	4	4	4

Metabolism of porphyrins (R-HSA-189445)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Down	Down	Up	Up
FDR	0.063	< 0.001	< 0.001	0.013
Av. FC	-0.777	-0.526	0.743	0.996
N Genes	22	22	22	22

Heme biosynthesis (R-HSA-189451)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Down	Down	Up	Up
FDR	0.118	0.006	0.001	0.025
Av. FC	-0.776	-0.473	0.784	1.09
N Genes	13	13	13	13

tRNA processing in the mitochondrion (R-HSA-6785470)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Up	Up	Down	Down
FDR	< 0.001	0.174	0.049	0.001
Av. FC	0.684	0.326	-0.472	-0.828
N Genes	23	23	23	23

Formation of the dystrophin-glycoprotein complex (DGC) (R-HSA-9913351)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Down	Up	Up	Up
FDR	0.016	0.941	0.778	0.016
Av. FC	-0.811	0.084	0.238	1.132
N Genes	19	19	19	19

Metabolism of Angiotensinogen to Angiotensins (R-HSA-2022377)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Down	Down	Up	Up
FDR	0.028	0.812	0.139	0.025
Av. FC	-2.132	-0.259	1.414	3.284
N Genes	9	9	9	9

Antimicrobial peptides (R-HSA-6803157)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Down	Down	Up	Up
FDR	0.025	0.518	0.074	0.028
Av. FC	-1.838	-0.343	0.94	2.439
N Genes	27	27	27	27

FASTK family proteins regulate processing and stability of mitochondrial RNAs (R-HSA-9837092)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Up	Up	Down	Down
FDR	0.005	0.614	0.074	0.002
Av. FC	0.53	0.263	-0.451	-0.718
N Genes	18	18	18	18

Complex I biogenesis (R-HSA-6799198)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Up	Up	Up	Down
FDR	< 0.001	0.904	0.807	0.002
Av. FC	0.293	0.076	-0.022	-0.238
N Genes	66	66	66	66

Mitochondrial RNA degradation (R-HSA-9836573)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Up	Up	Down	Down
FDR	0.005	0.486	0.074	0.003
Av. FC	0.446	0.233	-0.379	-0.59
N Genes	23	23	23	23

mRNA Splicing (R-HSA-72172)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Up	Up	Down	Down
FDR	< 0.001	0.177	0.916	0.008
Av. FC	0.204	0.083	-0.036	-0.155
N Genes	205	205	205	205

mRNA Splicing - Major Pathway (R-HSA-72163)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Up	Up	Down	Down
FDR	< 0.001	0.178	0.927	0.013
Av. FC	0.199	0.081	-0.033	-0.148
N Genes	197	197	197	197

Metabolism of RNA (R-HSA-8953854)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Up	Up	Down	Down
FDR	< 0.001	0.006	0.727	0.013
Av. FC	0.171	0.096	-0.052	-0.128
N Genes	683	683	683	682

Processing of Capped Intron-Containing Pre-mRNA (R-HSA-72203)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Up	Up	Down	Down
FDR	< 0.001	0.121	0.793	0.013
Av. FC	0.18	0.087	-0.048	-0.14
N Genes	271	271	271	271

HIV Transcription Initiation (R-HSA-167161)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Up	Up	Down	Down
FDR	< 0.001	0.551	0.802	0.025
Av. FC	0.216	0.095	-0.08	-0.2
N Genes	44	44	44	44

RNA Polymerase II HIV Promoter Escape (R-HSA-167162)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Up	Up	Down	Down
FDR	< 0.001	0.551	0.802	0.025
Av. FC	0.216	0.095	-0.08	-0.2
N Genes	44	44	44	44

RNA Polymerase II Promoter Escape (R-HSA-73776)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Up	Up	Down	Down
FDR	< 0.001	0.551	0.802	0.025
Av. FC	0.216	0.095	-0.08	-0.2
N Genes	44	44	44	44

RNA Polymerase II Transcription Pre-Initiation And Promoter Opening (R-HSA-73779)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Up	Up	Down	Down
FDR	< 0.001	0.551	0.802	0.025
Av. FC	0.216	0.095	-0.08	-0.2
N Genes	44	44	44	44

RNA Polymerase II Transcription Initiation (R-HSA-75953)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Up	Up	Down	Down
FDR	< 0.001	0.551	0.802	0.025
Av. FC	0.216	0.095	-0.08	-0.2
N Genes	44	44	44	44

RNA Polymerase II Transcription Initiation And Promoter Clearance (R-HSA-76042)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Up	Up	Down	Down
FDR	< 0.001	0.551	0.802	0.025
Av. FC	0.216	0.095	-0.08	-0.2
N Genes	44	44	44	44

Respiratory electron transport (R-HSA-611105)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Up	Up	Up	Down
FDR	< 0.001	0.764	0.485	0.028
Av. FC	0.243	0.066	0.007	-0.168
N Genes	149	149	149	149

tRNA processing (R-HSA-72306)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Up	Up	Down	Down
FDR	0.011	0.757	0.459	0.028
Av. FC	0.2	0.1	-0.136	-0.235
N Genes	118	118	118	118

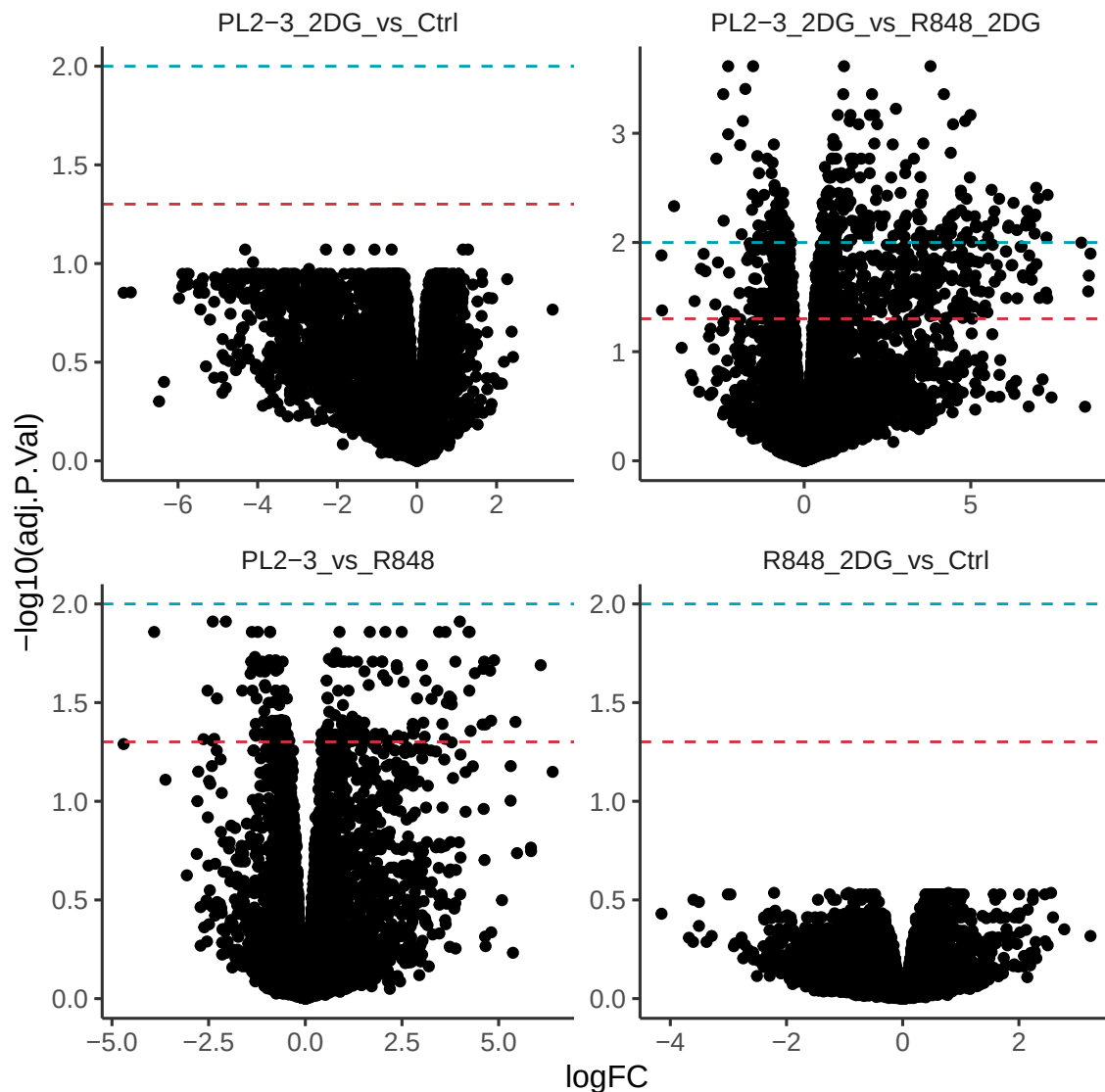
rRNA processing (R-HSA-72312)

	PL2-3_2DG_vs_Ctrl	R848_2DG_vs_Ctrl	PL2-3_vs_R848	PL2-3_2DG_vs_R848_2DG
Regulation	Up	Up	Down	Down
FDR	< 0.001	< 0.001	0.698	0.036
Av. FC	0.223	0.148	-0.061	-0.134
N Genes	199	199	199	199

Protein / Gene level results

Classical differential expression analysis performed on the genes / proteins of every dataset.

Note: Depending on the gene set analysis method used, the approach used to assess differential expression at the gene / protein level may vary to the approach used for the pathway level.



Top up-regulated protein/genes

PL2-3_2DG_vs_Ctrl

Identifier	logFC	AveExpr	t	P.Value	adj.P.Val	B
Hspb11	1.1549286	2.654251	5.355114	0.0000514	0.0849543	2.0739599
Cmc4	1.2847487	2.526511	5.295981	0.0000581	0.0849543	1.9648201
Sorbs2	1.0562661	1.081802	4.517153	0.0003000	0.1128166	0.4727490
Cibar1	0.7494709	2.847000	4.459152	0.0003398	0.1128166	0.2269731
Stac2	0.6653770	2.541697	4.395746	0.0003894	0.1128166	0.1062546
Zbtb4	0.5797626	4.148651	4.451726	0.0003453	0.1128166	0.0314824
Anp32a	0.7192529	3.537200	4.281839	0.0004979	0.1128166	-0.2090658
Cfap36	0.6265159	4.972001	4.323648	0.0004549	0.1128166	-0.2824710

Identifier	logFC	AveExpr	t	P.Value	adj.P.Val	B
Lamp3	0.8495443	1.222778	4.115085	0.0007144	0.1128166	-0.3472784
Fdx1	0.4921528	3.974522	4.172913	0.0006302	0.1128166	-0.4925424

R848_2DG_vs_Ctrl

Identifier	logFC	AveExpr	t	P.Value	adj.P.Val	B
Pdcd4	0.7879554	7.1581854	5.018862	0.0000994	0.2911714	1.4968533
Gpr156	2.5540597	-0.3933598	4.957579	0.0001132	0.2911714	-0.7398602
Ythdf2	0.4250090	7.2401744	4.830936	0.0001482	0.2963446	1.1193062
P2ry10	0.7012952	5.6716168	4.600827	0.0002427	0.2963446	0.6634814
Cyp39a1	0.7546005	3.9867163	4.407952	0.0003683	0.2963446	0.2996678
Dync2h1	0.6522602	3.0698588	4.336782	0.0004298	0.2963446	0.1528515
Calcr1	0.8562293	2.9965117	4.349203	0.0004183	0.2963446	0.1281292
Hpgds	1.6814923	2.7660952	4.292461	0.0004732	0.2963446	0.0176972
Scd1	0.3909964	9.6988996	4.282677	0.0004834	0.2963446	0.0011788
Shld3	0.7412431	1.9964439	4.295826	0.0004698	0.2963446	-0.0315741

PL2-3_vs_R848

Identifier	logFC	AveExpr	t	P.Value	adj.P.Val	B
C3	3.9961152	1.3344450	6.776223	2.90e-06	0.0122794	4.820796
Itgam	2.4927940	2.0724923	6.391368	6.10e-06	0.0138626	3.978072
Colec12	3.6252998	0.0110797	6.152835	9.70e-06	0.0138626	3.642391
Hk2	0.8855609	3.6503861	5.921048	1.55e-05	0.0138626	3.239123
Gngt2	1.6685328	5.4571972	5.853933	1.77e-05	0.0138626	2.995599
Slc43a1	2.0705087	4.9154926	5.812164	1.92e-05	0.0138626	2.958170
Lilra5	4.2492175	-0.3516176	5.756100	2.16e-05	0.0138626	2.918315
Mmp14	3.4627496	-0.8445943	6.247951	8.10e-06	0.0138626	2.634493
Cdc42ep2	4.2240366	-0.2951469	5.855409	1.76e-05	0.0138626	2.439808
Stx11	0.8013149	5.5723728	5.605296	2.93e-05	0.0177421	2.481162

PL2-3_2DG_vs_R848_2DG

Identifier	logFC	AveExpr	t	P.Value	adj.P.Val	B
Slc43a1	3.791460	4.915506	9.568424	0e+00	0.0002432	9.359820
Scd2	1.195754	7.126462	8.784776	1e-07	0.0002432	8.118823

Identifier	logFC	AveExpr	t	P.Value	adj.P.Val	B
Bcl2l11	1.173891	6.019625	8.158376	3e-07	0.0004386	7.091731
Hnf1b	4.194296	4.034782	8.009279	3e-07	0.0004386	6.913903
Cdh17	2.035531	7.004243	7.987276	4e-07	0.0004386	6.788642
Tmed6	2.753985	3.821791	7.705014	6e-07	0.0005967	6.396863
Vangl2	2.105753	3.253541	7.424653	1e-06	0.0006802	5.927370
Dpp4	1.391909	7.731960	7.497524	8e-07	0.0006802	5.923910
Pck2	1.011531	6.976206	7.422057	1e-06	0.0006802	5.790586
Hsd11b1	1.973619	6.507280	7.401831	1e-06	0.0006802	5.761088

Top down-regulated protein/genes
PL2-3_2DG_vs_Ctrl

Identifier	logFC	AveExpr	t	P.Value	adj.P.Val	B
Hid1	-1.7036975	3.0330346	-5.546399	0.0000347	0.0849543	2.4621592
Igf2r	-1.0646640	4.4458049	-5.533884	0.0000356	0.0849543	2.3528566
Als2	-0.6357450	5.2121953	-5.512327	0.0000372	0.0849543	2.2423184
Slc43a1	-2.2852227	4.9155333	-5.426324	0.0000444	0.0849543	2.1869804
Itgad	-4.3130887	0.7887752	-5.654009	0.0000279	0.0849543	1.9496709
Tnfsf14	-4.1141447	-0.8981660	-5.160476	0.0000769	0.0984710	0.1846450
Trim47	-2.7103361	-1.6101155	-5.066595	0.0000936	0.1064837	0.3208772
Sgpl1	-0.5059618	7.6557763	-4.985385	0.0001110	0.1128166	1.0668316
Mrc1	-2.8119288	0.4019860	-4.871726	0.0001410	0.1128166	1.0616512
Man1b1	-0.4882318	6.1385270	-4.933839	0.0001237	0.1128166	0.9766768

R848_2DG_vs_Ctrl

Identifier	logFC	AveExpr	t	P.Value	adj.P.Val	B
Naprt	-0.9266452	2.739189	-5.063147	0.0000905	0.2911714	1.4296180
Trim47	-2.2143132	-1.610156	-5.072180	0.0000888	0.2911714	-0.9440623
Slc16a6	-0.7541650	5.113030	-4.511135	0.0002946	0.2963446	0.4885672
Clk3	-0.4840962	6.691717	-4.329071	0.0004370	0.2963446	0.0975386
Ccdc88b	-0.7215004	4.658874	-4.196757	0.0005829	0.2963446	-0.1312984
Pdk1	-0.5564480	4.342906	-4.159822	0.0006318	0.2963446	-0.2022091
Lilra6	-0.8732917	3.675478	-4.011401	0.0008738	0.2963446	-0.4802666
Cyb5r1	-0.5002838	4.641066	-3.893409	0.0011313	0.2963446	-0.7147810
Tmem86b	-0.8889005	2.210686	-3.917613	0.0010729	0.2963446	-0.7602336

Identifier	logFC	AveExpr	t	P.Value	adj.P.Val	B
Dennd6b	-0.9695515	3.701659	-3.865606	0.0012023	0.2963446	-0.7622546

PL2-3_vs_R848

Identifier	logFC	AveExpr	t	P.Value	adj.P.Val	B
Lamp3	-2.0549721	1.2227373	-6.789304	2.80e-06	0.0122794	4.218226
Lurap1	-2.3923843	0.7412013	-6.668037	3.60e-06	0.0122794	3.785930
Akt3	-1.2291080	3.7164890	-5.935474	1.50e-05	0.0138626	3.265818
Cr2	-1.3830363	9.2833905	-5.893425	1.63e-05	0.0138626	3.033822
Fchsd2	-0.9075838	6.8368807	-5.785890	2.03e-05	0.0138626	2.823146
B3gnt5	-0.9132449	7.3146830	-5.777089	2.07e-05	0.0138626	2.802919
Gpr156	-3.9087876	-0.3933597	-5.959962	1.43e-05	0.0138626	1.410989
Cacna1d	-1.3027346	1.5101037	-5.555397	3.25e-05	0.0185619	2.427012
Usp53	-0.9280258	3.5819713	-5.464176	3.92e-05	0.0192889	2.352514
Brwd1	-0.7667979	6.6538171	-5.439384	4.12e-05	0.0192889	2.117890

PL2-3_2DG_vs_R848_2DG

Identifier	logFC	AveExpr	t	P.Value	adj.P.Val	B
Stac2	-1.531583	2.5416692	-9.005844	1.0e-07	0.0002432	8.419892
Lamp3	-2.280127	1.2227502	-8.960396	1.0e-07	0.0002432	7.880531
Ddx25	-1.764389	1.9957269	-8.355908	2.0e-07	0.0003928	7.309139
Serpina7	-2.428008	0.5645002	-7.943280	4.0e-07	0.0004386	6.203831
Rgs18	-1.838297	3.9499503	-7.287423	1.2e-06	0.0007726	5.639964
Lurap1	-2.279023	0.7412142	-6.943837	2.3e-06	0.0010211	4.619846
Zbtb4	-0.910097	4.1486230	-6.737436	3.4e-06	0.0012656	4.613421
Cd55	-1.918134	8.1568539	-6.695626	3.6e-06	0.0012823	4.437388
Vpreb3	-1.405477	6.0312551	-6.507566	5.2e-06	0.0016171	4.096998
Trdmt1	-1.110412	2.8719132	-6.353390	7.0e-06	0.0017089	3.992646